Stats Under the Stars 9 - Forecasting delle segnalazioni sulla rete stradale

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1 Problem Formulation

The project addressed the prediction of monthly incident counts for each street during 2025. Two substantially different formulations of the task were considered. The first formulation treated the challenge as a direct forecasting problem. Historical incidents, temporal variables, street identifiers, and road metadata were used to predict every future street–month count. Under this formulation, the model had to estimate both the annual number of incidents and their distribution across the twelve months. The second formulation was developed after identifying annual incident information for 2024 and 2025 in the metadata. The relevant variables contained annual counts disaggregated by incident type. Their sum provided the annual total for each street and target year. Consequently, the task was reformulated as the allocation of a known annual total across twelve months. This distinction was decisive. Direct monthly forecasting obtained a public RMSLE of 0.69483, whereas the final annual-allocation ensemble obtained 0.47927.

2 Data Preparation and Validation

The historical dataset contained incident-level observations from 2016 to 2023. Registration dates were converted into monthly timestamps, street identifiers were standardized, and exact duplicate records were removed. Monthly targets were constructed by aggregating incidents by street, year, and month. Street-level metadata were merged with the monthly data. The available variables included the number of road sections, municipalities and provinces crossed, minimum and maximum kilometre markers, average road coverage, and average traffic for light and heavy vehicles. Additional features included estimated road length, total traffic, and the share of heavy traffic. The year 2023 was used for local validation because it was the latest complete historical year and the closest available approximation to forecasting 2024 and 2025. Models were trained using information from 2016 to 2022 and evaluated on 2023. This chronological split prevented future target information from entering the validation features. For the allocation models, the 2023 annual totals and incident-type composition were used in the same way as the corresponding information for 2024 and 2025. The validation notebook generated 5,568 possible street–month combinations for 2023. However, the reported local RMSLE was calculated on the 2,463 combinations for which an observed monthly target was present in the grouped historical table. Combinations without an observed incident record remained missing rather than being explicitly converted to zero. Therefore, local validation was used primarily for model comparison and ensemble-weight selection, while the public leaderboard provided the definitive evaluation on the complete test structure.

3 Street Coverage and Hierarchical Information

The streets in the test set did not have the same historical coverage. The modelling strategy therefore distinguished three groups.

1. **Streets with direct historical information.** Among the 627 test streets, 491 had an exact normalized match in the training data. For these streets, street-specific historical statistics could be constructed.

2. **Unseen streets with an observed street family.** A further 30 streets had no exact historical match, but their broader street family appeared in the training data. These cases could use family-level historical information.
3. **Unseen streets without an observed street family.** The remaining 106 streets had neither an exact historical match nor a street family represented in the training data. Their predictions relied on calendar effects, global historical patterns, target-year incident composition, and road metadata.

The street family was extracted from the initial component of the street code. This representation allowed information to be shared between related streets. Two indicators, `road_has_history` and `family_has_history`, identified the information available for each prediction. The resulting feature hierarchy used street-level statistics when direct history existed, family-level statistics when only the broader family was observed, and global, seasonal, incident-type, and metadata variables when no related historical information was available. Road metadata were particularly important for unseen streets. A metadata record was available for every test street. Numerical road characteristics were available for 250 of the 491 streets with direct history and for all 136 streets without an exact historical match. Missing numerical values were retained as missing and handled directly by CatBoost.

4 Direct Monthly Forecasting

The first solution treated each street–month count as an unrestricted prediction target. CatBoost and LightGBM models were trained using categorical street variables, calendar information, historical incident statistics, and physical road characteristics. CatBoost was used to model nonlinear interactions involving categorical street identifiers, while LightGBM provided a complementary gradient-boosting specification. Historical features were calculated only from periods preceding the validation year. This approach had to solve two prediction problems simultaneously: estimating the annual volume of incidents for each street and determining their monthly timing. Furthermore, the twelve monthly predictions were not constrained to sum to a known annual value. Errors in the annual level therefore affected all monthly predictions for the same street. Using 2023 as the temporal validation year, the retained local result was approximately 0.712 RMSLE. The corresponding public leaderboard score was 0.69483.

5 Reformulation as an Allocation Problem

The metadata contained 16 incident-type variables for 2024 and 17 for 2025. Summing these variables produced the annual incident total for each street and target year. The annual total therefore did not need to be forecast. The remaining task consisted of estimating monthly proportions satisfying

$$\sum_{m=1}^{12} p_{r,y,m} = 1,$$

where $p_{r,y,m}$ is the proportion of annual incidents assigned to month m for street r and year y. Monthly predictions were obtained from

$$\widehat{Y}_{r,y,m} = A_{r,y}\widehat{p}_{r,y,m},$$

where $A_{r,y}$ is the annual total obtained from the metadata. This formulation removed uncertainty about the annual magnitude and concentrated the model on the temporal distribution of a known total. sectionAllocation Models

5.1 CatBoost with Incident-Type Composition

The main allocation model used the street identifier, street family, street type, year, calendar month, annual total, annual counts by incident type, incident-type shares, historical coverage indicators, and road metadata. CatBoost generated a non-negative raw score for each month. The twelve scores were normalized within each street–year group:

$$\widehat{p}^{B}_{r,y,m} = \frac{\widehat{z}_{r,y,m}}{\sum_{j=1}^{12} \widehat{z}_{r,y,j}}.$$

The normalized outputs represented monthly allocation shares. This model obtained a local RMSLE of 0.49701 and a public RMSLE of 0.48319.

5.2 Historical Seasonality by Incident Type

The second component estimated a monthly distribution for each incident type using historical data from 2016 to 2022. Add-one smoothing was applied so that every month received a positive probability. For each street–year, the annual count of each incident type was distributed according to its historical monthly profile. The type-specific allocations were then summed to obtain the total monthly allocation. This model obtained a local RMSLE of 0.49912. The component was simpler than CatBoost but provided a stable seasonal pattern directly linked to the annual incident composition.

6 Final Ensemble and Calibration

Different model combinations were compared using the 2023 validation period. The best local result was obtained with

$$\widehat{p}_{r,y,m} = 0.75\widehat{p}^{B}_{r,y,m} + 0.25\widehat{p}^{C}_{r,y,m},$$

where B denotes the CatBoost allocation and C denotes historical incident-type seasonality. The combined shares were normalized again within each street–year group. The ensemble obtained a local RMSLE of 0.49676. Continuous monthly predictions were converted into integer counts using the largest-remainder method. The integer part of each prediction was assigned first, and the remaining incidents were allocated to the months with the largest fractional remainders. This procedure guaranteed that all predictions were non-negative integers, every street–year contained twelve monthly predictions, and the monthly predictions summed exactly to the annual total. The final submission contained 15,048 predictions and no street–year with an incorrect annual sum.

7 Results and Key Design Principles

Approach	Local RMSLE	Public RMSLE
Direct monthly forecasting	0.712	0.694
Annual-calibrated CatBoost	–	0.485
CatBoost with incident-type composition	0.497	0.483
Historical incident-type seasonality	0.499	–
Final 75/25 allocation ensemble	**0.496**	**0.479**

The final ensemble reduced the public RMSLE from 0.69483 to 0.47927, corresponding to an improvement of approximately 31.0%. The main contribution was not simply the replacement of one predictive algorithm with another. The decisive improvement resulted from solving a different statistical problem. Direct forecasting estimated both annual magnitude and monthly timing, whereas the final method used the annual totals already available in the metadata and estimated only their monthly distribution. A second design principle was the hierarchical treatment of street information. Direct street history was used when available, family-level history was used for unseen streets belonging to observed families, and metadata, seasonal, and global information supported streets without historical coverage. A third principle was the use of metadata as structural information rather than only as conventional model features. The incident-type variables determined the annual totals and also described their composition. Finally, the solution combined a flexible machine-learning model with a stable deterministic seasonal component. CatBoost captured nonlinear relationships between streets, months, metadata, and incident composition, while historical seasonality reduced instability. Exact annual calibration ensured that every prediction remained consistent with the information provided in the dataset.